Mission:- Access Local DB Data into cloud apex.oracle.com

Phase 1 — Install ORDS for Oracle 11g XE (FOUNDATION)

Important rule

For **11g XE**, do **NOT** use latest ORDS.
Use **ORDS 18.x** (stable & compatible).

STEP 1 — Prerequisites (must match exactly)

Java

- Install **Java 8 (JDK 8)**
  Check:

java –version

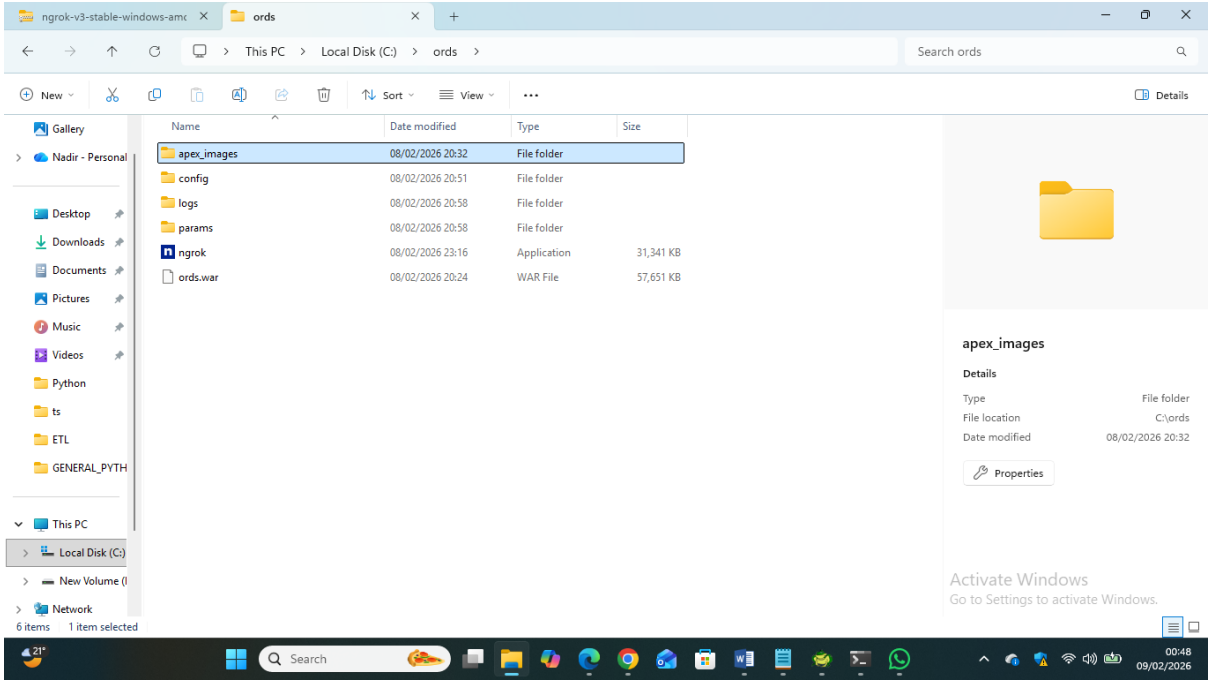
Must show 1.8

✅ Folder structure (example) – Manual Creation

C:\ords\

```
├── ords.war
├── config\
└── logs\
```

Additional shows in snap



STEP 2:- Prepare Oracle 11g XE

Connect as SYS:

ALTER SYSTEM SET dispatchers='(PROTOCOL=TCP)(SERVICE=xeXDB)' SCOPE=BOTH;

ALTER SYSTEM SET local_listener='(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521))' SCOPE=BOTH;

STEP 3 — Configure ORDS
cd C:\ords

java -jar ords.war install

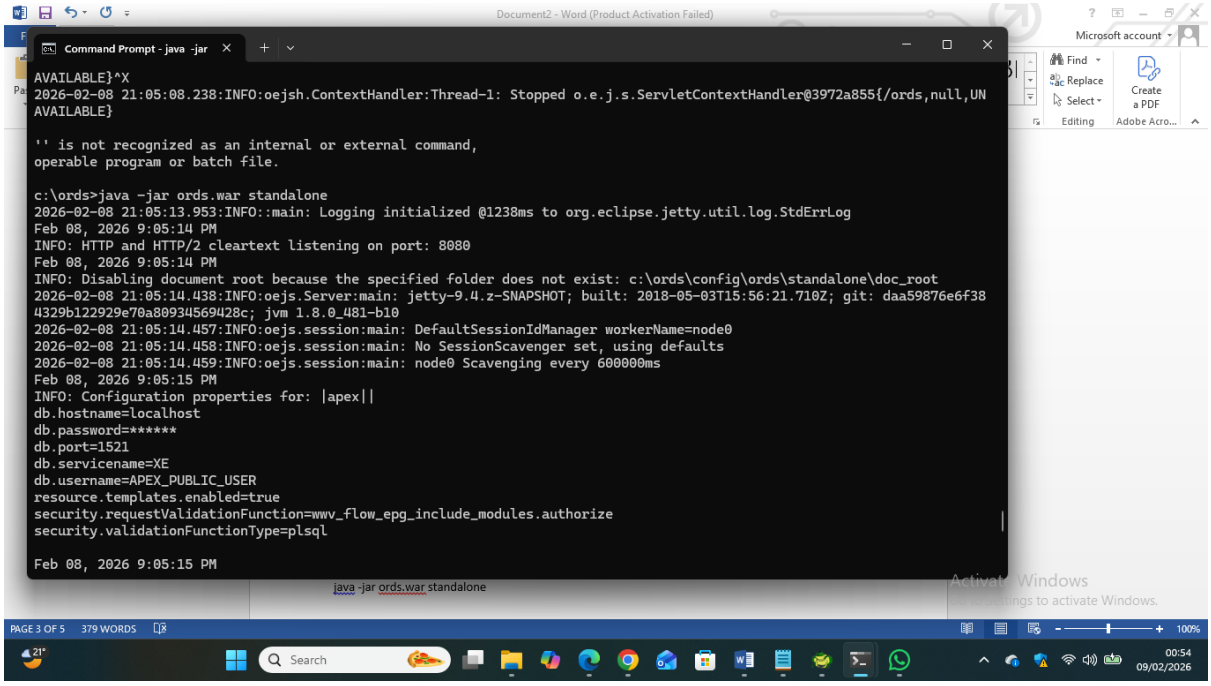
Answer prompts carefully:

Question	Value
Enter database hostname	localhost
Enter database port	1521
Enter database service name	XE
SYS user password	(your SYS password)
Create ORDS user	YES
Create APEX_PUBLIC_USER	YES (even if APEX not used locally)
Enable SQL Developer Web	NO

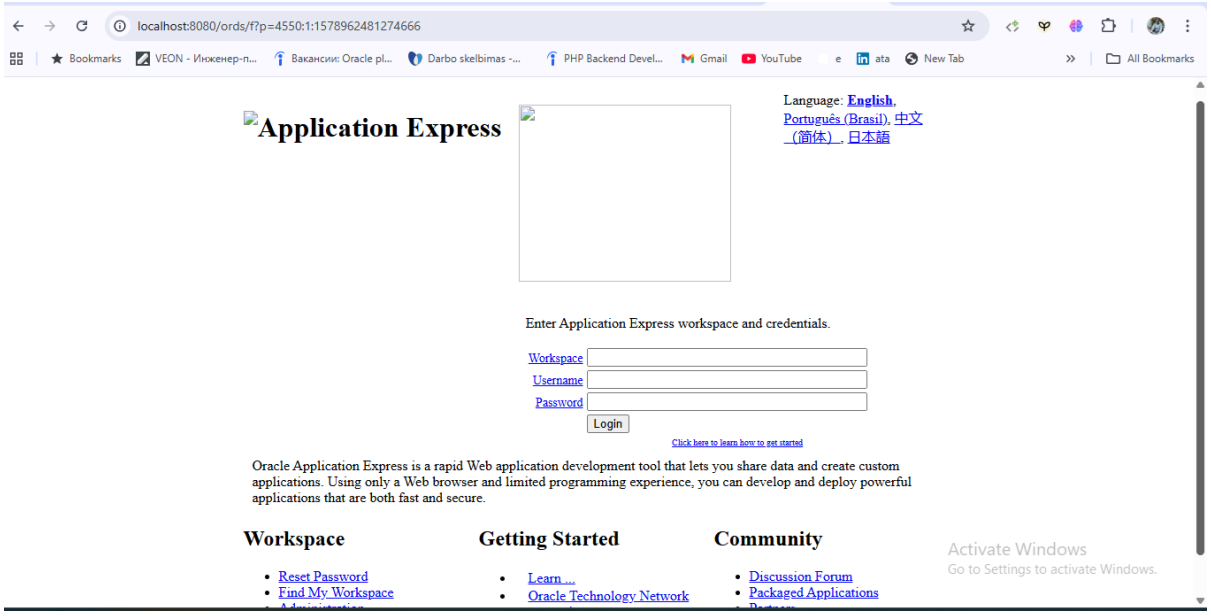
✓ ORDS schema will be created
✓ REST support enabled

STEP 4 — Start ORDS

java -jar ords.war standalone



http://localhost:8080/ords/



Note:- This launching without error is required, if this is not done nothing will be happen. So first this page has to be launched no matter the login will happen or not but this has to be generated successfully like it showed

Phase 2 — Make ORDS reachable from APEX Cloud

This is the **most important part**.

❌ What will NOT work

- localhost
- 127.0.0.1
- 192.168.x.x

✅ What WILL work

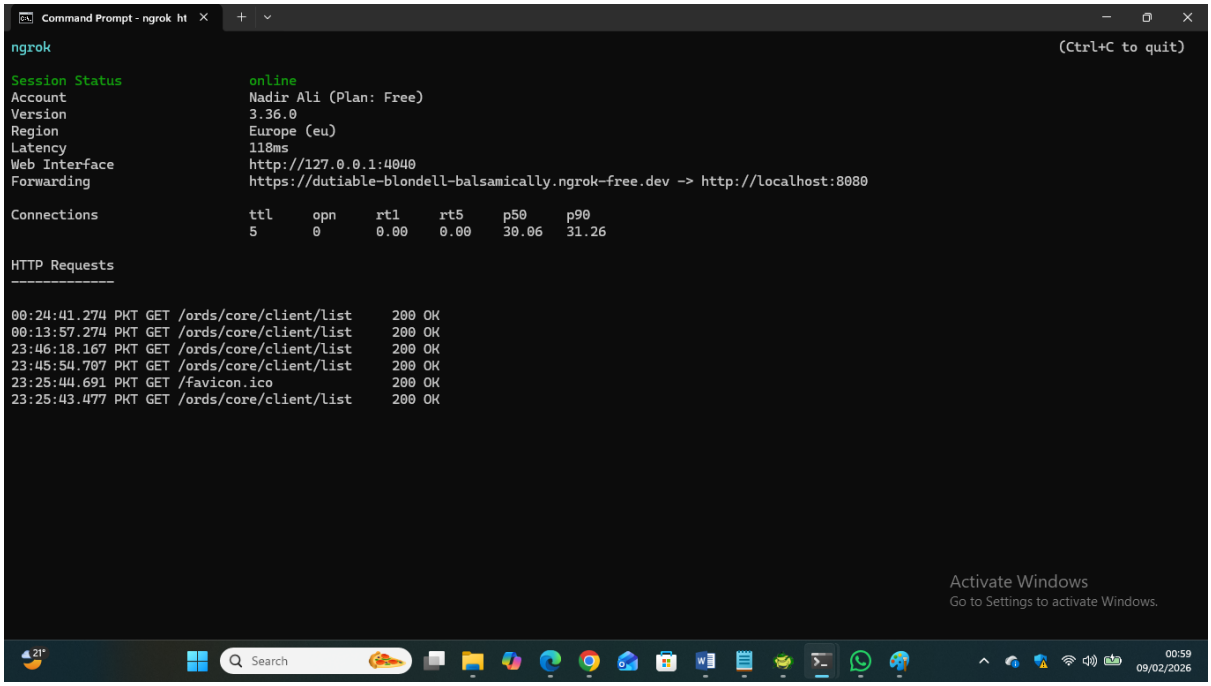
One of these:

Option A (Quick test)

- Use **ngrok** (HTTP tunnel)

On CMd type following

ngrok http 8080



Phase 3 — Create FIRST REST API (Simple & Safe)

Run in your schema:

```
BEGIN
  ORDS.ENABLE_SCHEMA(
```

```
    p_enabled => TRUE,
    p_schema  => 'YOUR_SCHEMA',
    p_url_mapping_type => 'BASE_PATH',
    p_url_mapping_pattern => 'core',
    p_auto_rest_auth => FALSE
  );
END;
/
```

Create test REST:

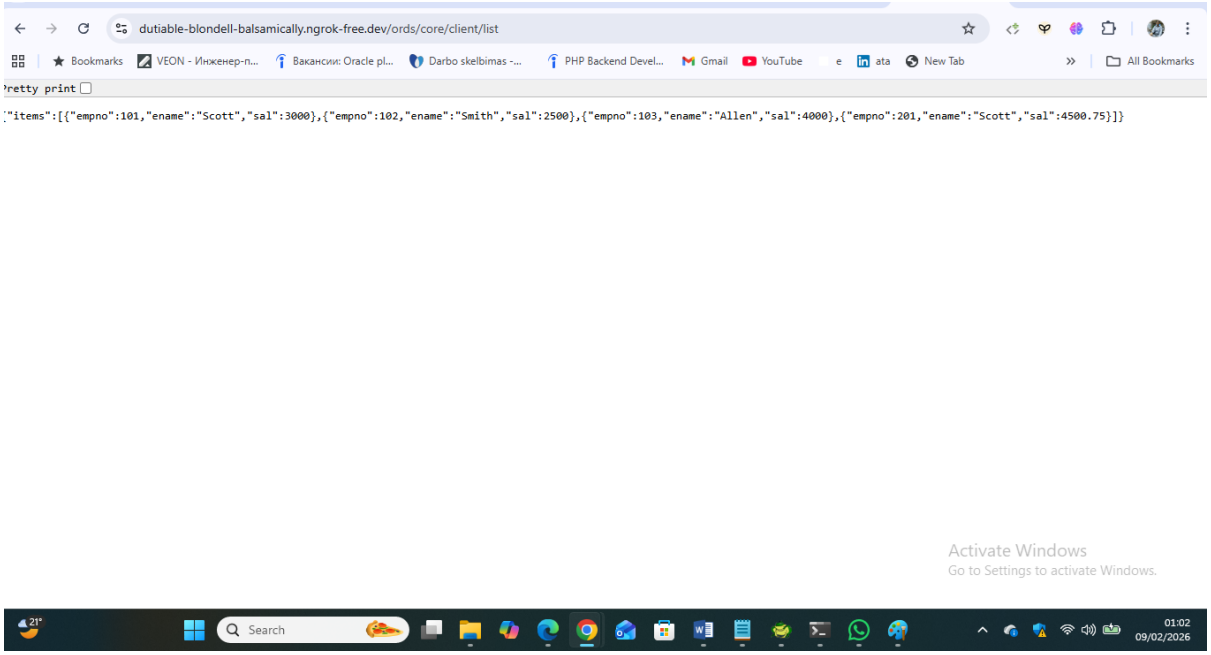
```
BEGIN
  ORDS.DEFINE_MODULE(
    p_module_name => 'hello_api',
    p_base_path   => '/hello/',
    p_items_per_page => 0
  );

  ORDS.DEFINE_TEMPLATE(
    p_module_name => 'hello_api',
    p_pattern     => 'data'
  );

  ORDS.DEFINE_HANDLER(
    p_module_name => 'hello_api',
    p_pattern     => 'data',
    p_method      => 'GET',
    p_source_type => ORDS.SOURCE_TYPE_QUERY,
    p_source      => 'SELECT "Hello from 11g XE" msg FROM dual'
  );

  COMMIT;
```

END;

/

https://xxxx.ngrok.io/ords/core/hello/data

If JSON appears → **YOU WON** 🎉



Phase 4 — Consume in APEX (apex.oracle.com)

- Shared Components
- Web Source Modules
- Create from REST
- Paste ngrok/public URL
- Create Report

Do NOT jump ahead

Ignore AI, prompts, metadata for now.

Success criteria for this phase:

✓ ORDS running
✓ REST URL reachable from internet
✓ JSON visible
✓ APEX report shows data

You have **Oracle 11g XE**, so **ORDS version choice is NOT flexible**.

✅ ORDS ↔ Oracle Database Compatibility (FACTS)

● **What will NOT work**

- ORDS **19.x**
- ORDS **20.x**
- ORDS **21.x**
- ORDS **22.x**
- ORDS **23.x**

These require **Oracle 12c+**
They will **fail silently or break during install** on 11g.

✅ What WILL work with Oracle 11g XE

✓ Recommended (Safe & Proven)

ORDS 18.4.x

This is the **last stable ORDS line** that works properly with:

- Oracle 11g
- Oracle 11g XE
- Java 8

This is what you should use. No debate here.

Java compatibility (also critical)

ORDS Version	Java Required
ORDS 18.x	**Java 8 (ONLY)**
ORDS 19+	Java 11+

What I strongly recommend for you

Use this exact combo:

- **Oracle 11g XE**
- **ORDS 18.4**
- **Java JDK 8**

This combo is:

- Stable
- Predictable
- Works perfectly for REST
- Enough for your APEX + Prompt-AI goal

Common mistake (avoid this)

❌ Downloading “latest ORDS”
❌ Running with Java 11
❌ Assuming backward compatibility

Oracle is very strict here.

From command prompt:

Prerequisite:-

Following commands on SYS level

ALTER USER APEX_PUBLIC_USER IDENTIFIED BY ORDS123 ACCOUNT UNLOCK;

ALTER USER APEX_LISTENER IDENTIFIED BY ORDS123 ACCOUNT UNLOCK;

ALTER USER APEX_REST_PUBLIC_USER IDENTIFIED BY ORDS123 ACCOUNT UNLOCK;

--- Alternate if not

CREATE USER APEX_LISTENER IDENTIFIED BY ORDS123 ACCOUNT UNLOCK;

CREATE USER APEX_REST_PUBLIC_USER IDENTIFIED BY ORDS123 ACCOUNT UNLOCK;

Step 1 — Make local XE database REST-accessible via ORDS

You already installed ORDS 18.4 and it is running. Now we need **a REST endpoint for a simple table**.

1.1 Pick a table for testing

Example table (if not already exists):

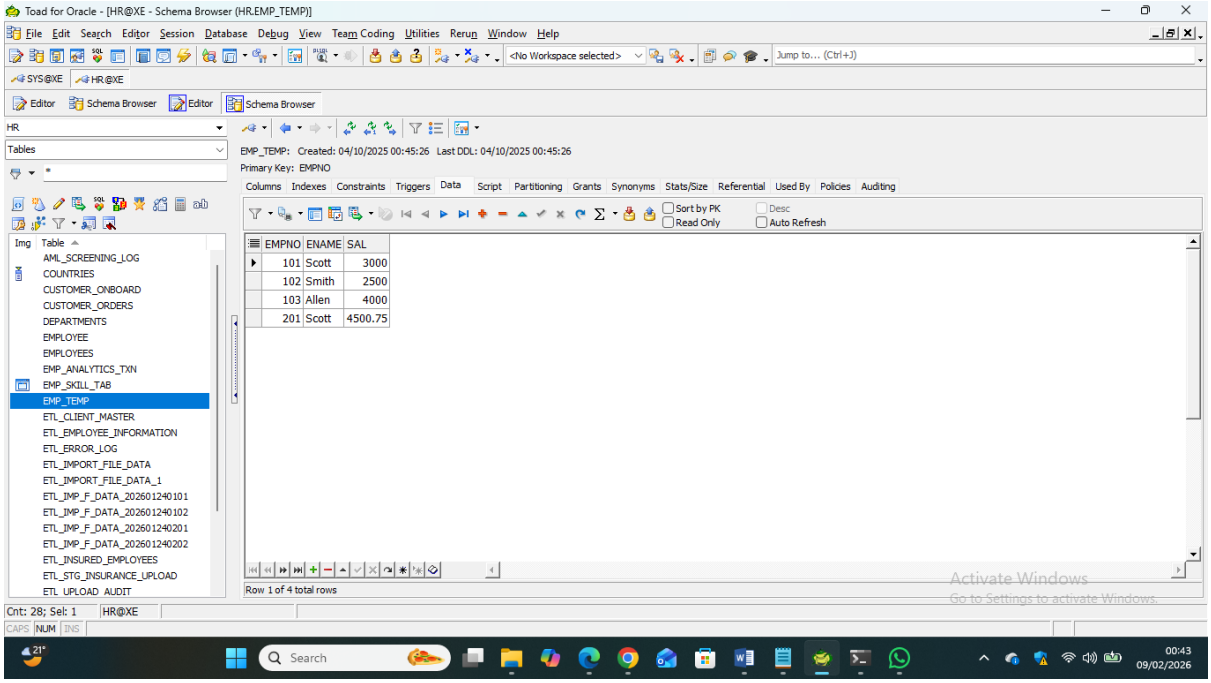
```
CREATE TABLE client_master (
   client_id   NUMBER,
   client_name VARCHAR2(100),
   city        VARCHAR2(50)
);
```

INSERT INTO client_master VALUES (1, 'ABC Corp', 'Karachi');

INSERT INTO client_master VALUES (2, 'XYZ Ltd', 'Lahore');

COMMIT;

1.2 Enable your schema for REST -- Schema that hold data to be displayed



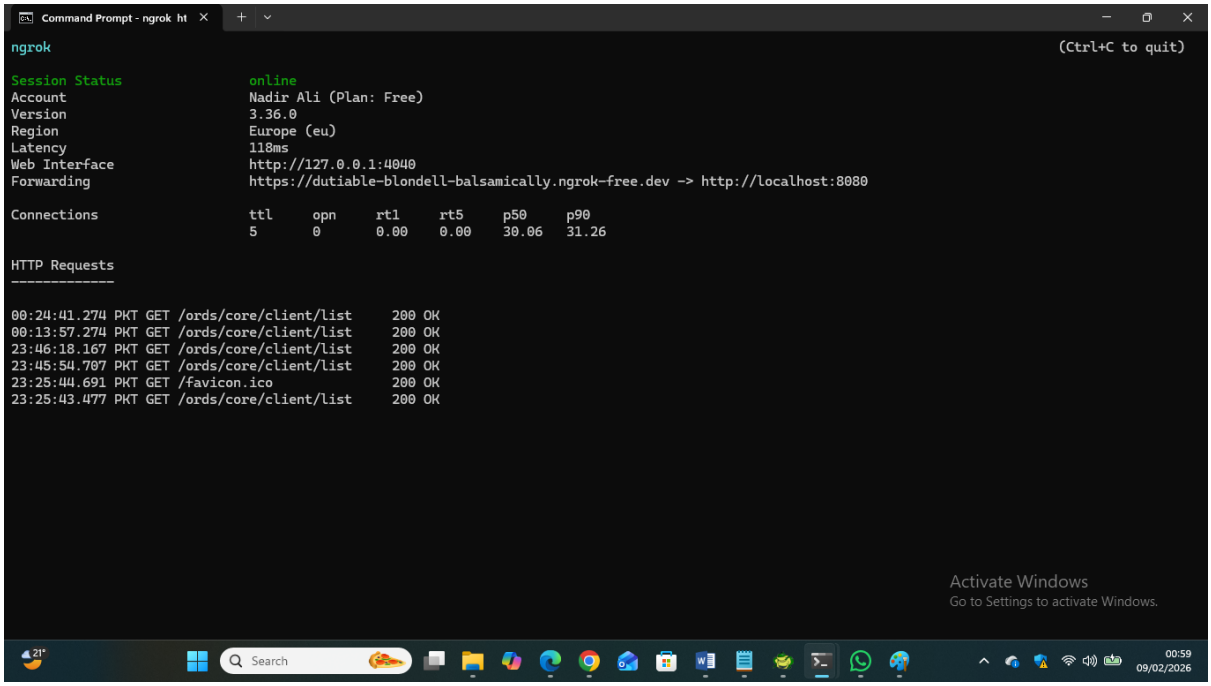

```
BEGIN
  ORDS.ENABLE_SCHEMA(
    p_enabled => TRUE,
    p_schema  => 'YOUR_SCHEMA', -- replace with your DB user
    p_url_mapping_type => 'BASE_PATH',
    p_url_mapping_pattern => 'core',
    p_auto_rest_auth => FALSE
  );
END;
/
```

This ensures **ORDS can expose REST services** for this schema.

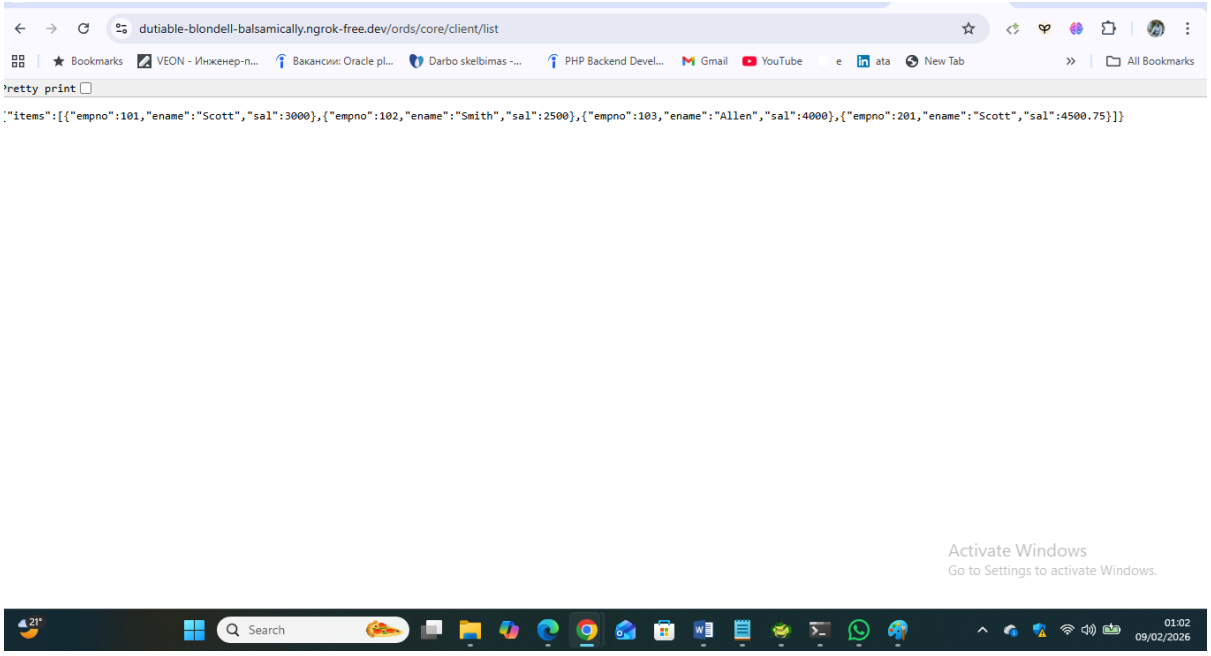
1.3 Create a simple REST module

```
BEGIN
  ORDS.DEFINE_MODULE(
    p_module_name => 'client_api',
    p_base_path   => '/client/',
    p_items_per_page => 0
  );

  ORDS.DEFINE_TEMPLATE(
    p_module_name => 'client_api',
    p_pattern     => 'list'
  );

  ORDS.DEFINE_HANDLER(
    p_module_name => 'client_api',
    p_pattern     => 'list',
    p_method      => 'GET',
    p_source_type => ORDS.SOURCE_TYPE_QUERY,
    p_source      => 'SELECT EMPNO, ENAME, SAL FROM EMP_TEMP'
  );

  COMMIT;
END;
/
```

Re-testing with real scenario



Generated forwarding https://dutiable-blondell-balsamically.ngrok-free.dev/ords/core/client/list



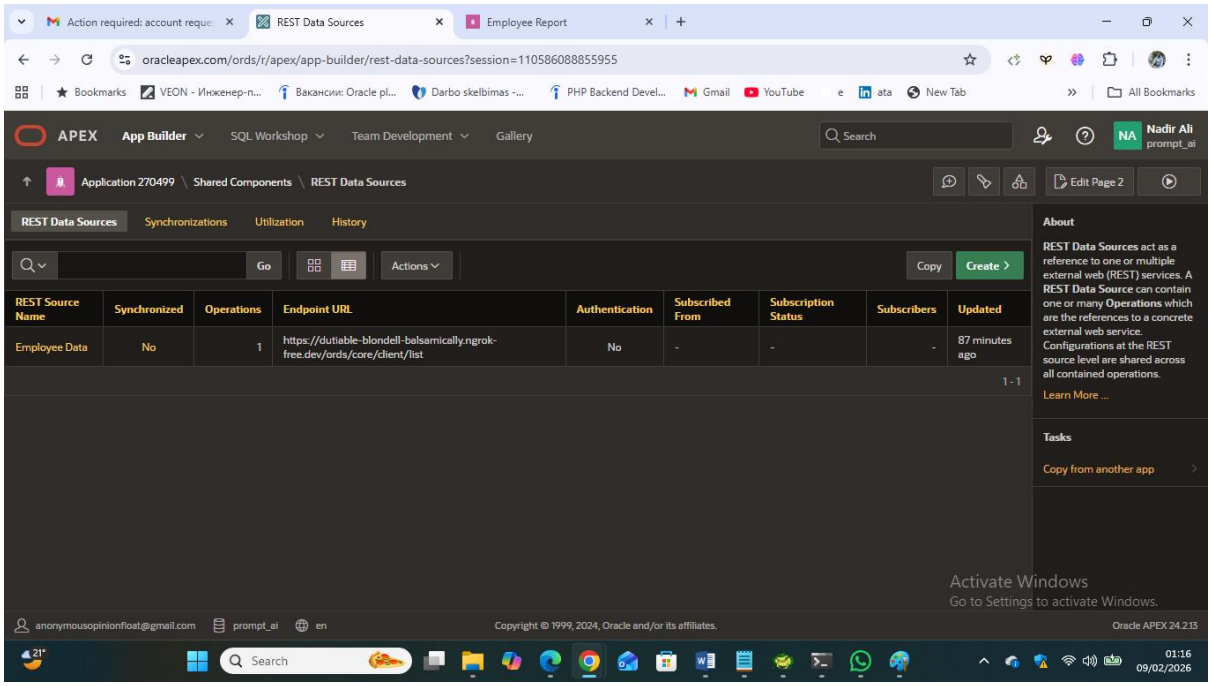
JSON result tested. Perfect now.

Correct Steps to Connect Local XE REST to APEX Cloud

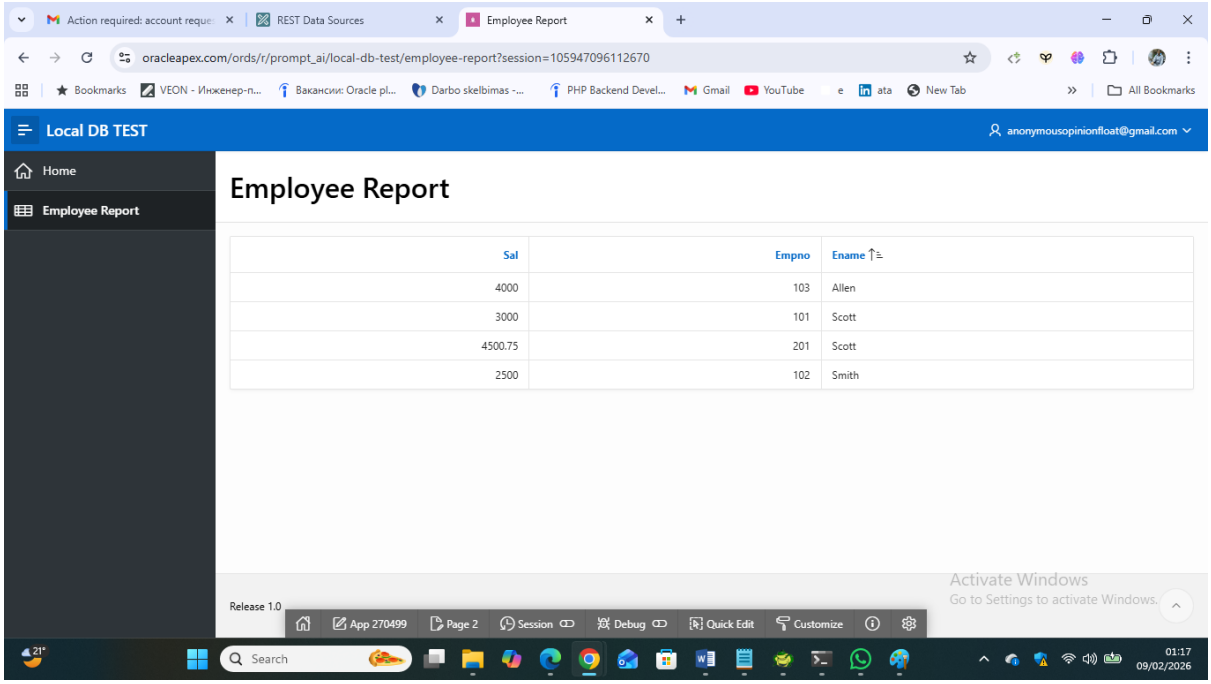
1. **Login** to your apex.oracle.com workspace.
2. **Create a new Application** (blank or minimal) first:
   - Go to **App Builder → Create → New Application**
   - Give it a name (e.g., Local DB Test)
   - Finish creation (no pages required yet).
3. **Open the Application**.
4. Go to **Shared Components → Web Source Modules → Create**.
5. Choose **REST URL** as the source type.
6. Enter your **publicly accessible REST URL** (from ngrok or port-forwarded ORDS):

https://dutiable-blondell-balsamically.ngrok-free.dev/ords/core/client/list

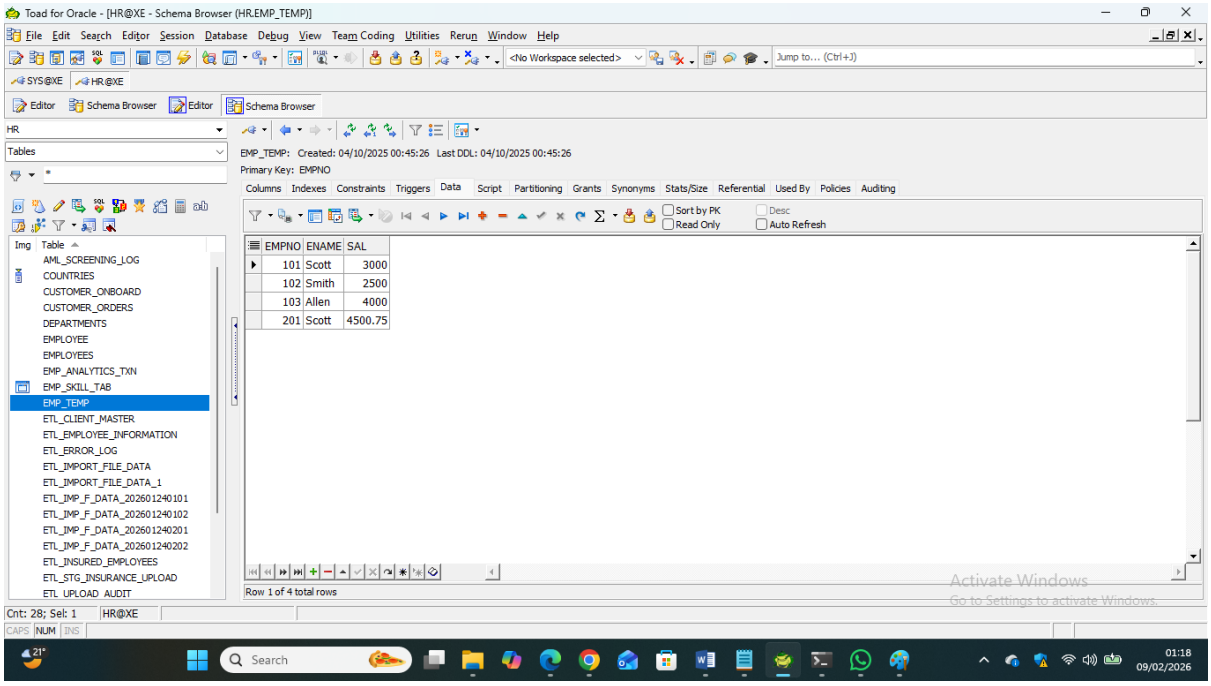
7. ☐ Method: GET
8. ☐ Test the connection → APEX should detect columns from JSON (CLIENT_ID, CLIENT_NAME, CITY).
9. ☐ Finish setup → **create a report page** using the Web Source Module.
10. Run the application → You should now see **data from your local XE database** in the cloud APEX report.



Local Cloud on Data


Local DB TEST
anonymousopinionfloat@gmail.com
Home
Employee Report
Employee Report
Sal
Empno
Ename
4000 103 Allen
3000 101 Scott
4500.75 201 Scott
2500 102 Smith
Release 1.0
App 270499
Page 2
Session
Debug
Quick Edit
Customize

On-Prem Data


Toad for Oracle - [HR@XE - Schema Browser (HR.EMP_TEMP)]
EMP_TEMP: Created: 04/10/2025 00:45:26 Last DDL: 04/10/2025 00:45:26
Primary Key: EMPNO
EMPNO ENAME SAL
101 Scott 3000
102 Smith 2500
103 Allen 4000
201 Scott 4500.75
Row 1 of 4 total rows